

Day 13: Logistic Regression

AI and Machine Learning Course

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Today's Agenda

What is Logistic Regression?

- A **classification algorithm** used to predict categorical outcomes (Yes/No, 0/1)
- Despite its name, it is used for **classification**, not regression
- Predicts the **probability** that an input belongs to a particular class
- Output is always between **0 and 1**

Examples of Use Cases

- Will a student be admitted? (Yes/No)
- Is an email spam? (Spam/Not Spam)
- Does a patient have a disease? (Positive/Negative)

The Sigmoid Function

Logistic Regression uses the **Sigmoid (Logistic) Function** to map any real number to a value between 0 and 1:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

where $z = b_0 + b_1x_1 + b_2x_2 + \cdots + b_nx_n$

- If $\sigma(z) \geq 0.5 \rightarrow$ Predict Class **1** (Positive)
- If $\sigma(z) < 0.5 \rightarrow$ Predict Class **0** (Negative)
- The value 0.5 is called the **decision threshold**

Linear Regression vs Logistic Regression

Feature	Linear Regression	Logistic Regression
Output	Continuous value	Probability (0 to 1)
Problem Type	Regression	Classification
Function	$y = mx + b$	$p = \frac{1}{1+e^{-(mx+b)}}$
Target Variable	Numerical	Categorical (0/1)
Metrics	MAE, MSE, RMSE, R^2	Accuracy, Precision, Recall, F1
Example	Predict house price	Predict admission

Key Difference

Linear Regression predicts a **quantity**. Logistic Regression predicts a **probability** which is then converted to a class label.

Confusion Matrix

A confusion matrix summarizes the prediction results of a classification model:

		Predicted	
		Negative (0)	Positive (1)
Actual	Negative (0)	TN	FP
	Positive (1)	FN	TP

- **TP (True Positive)**: Correctly predicted as positive
- **TN (True Negative)**: Correctly predicted as negative
- **FP (False Positive)**: Incorrectly predicted as positive (Type I Error)
- **FN (False Negative)**: Incorrectly predicted as negative (Type II Error)

Accuracy and Precision

Accuracy

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

- Proportion of **all correct predictions**
- Best when classes are balanced
- Can be misleading with imbalanced data

Precision

$$Precision = \frac{TP}{TP + FP}$$

- Of all predicted positives, how many are actually positive?
- Important when **false positives are costly** (e.g., spam filter)

Recall and F1 Score

Recall (Sensitivity)

$$Recall = \frac{TP}{TP + FN}$$

- Of all actual positives, how many did we correctly identify?
- Important when **false negatives are costly** (e.g., disease detection)

F1 Score

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

- Harmonic mean of Precision and Recall
- Balances both metrics into a single score
- Useful when you need a trade-off between Precision and Recall

When to Use Which Metric?

Metric	Use When
Accuracy	Classes are balanced and all errors matter equally
Precision	False positives are costly (don't want to wrongly flag something)
Recall	False negatives are costly (don't want to miss something important)
F1 Score	You need a balance between precision and recall, especially with imbalanced data

Tip

Always use **multiple metrics together** to get a complete picture of model performance.

Summary

- 1 Logistic Regression is a **classification algorithm** that predicts probabilities using the Sigmoid function
- 2 It outputs a value between 0 and 1, converted to class labels using a threshold (default 0.5)
- 3 The **Confusion Matrix** breaks down predictions into TP, TN, FP, FN
- 4 Key evaluation metrics:
 - **Accuracy** — overall correctness
 - **Precision** — quality of positive predictions
 - **Recall** — coverage of actual positives
 - **F1 Score** — balance between precision and recall
- 5 Choose metrics based on what type of error is more costly in your problem